# Gene-expression associations: source selection and target tests

14 of 20 source-selected associations had target statistical support in the same direction; 20 were testable. Support requires target BH q ≤ 0.05 across the full frozen family.

Saved public-cohort computation: GSE42568 source → GSE20711 target; exploratory ER association, not locked or clinical validation.

| Identity | Recorded value |
|---|---|
| Run ID | public_er_pass2 |
| Analysis | race |
| Evidence origin | public_computed |
| Evaluation role | exploratory |
| Independent provenance authenticated | No |

Research use only. No clinical certification or causal conclusion is established. Raw uploaded expression/metadata files are excluded; input hashes identify the originals.

## Study inputs and task eligibility

Recorded declarations, counts and task-specific restrictions; declarations are not acquisition audits.

| Recorded field | Source | Target |
|---|---|---|
| Cohort identity | GSE42568_ER | GSE20711_ER |
| Accession | GSE42568 | GSE20711 |
| Included samples | 101 | 87 |
| Features | 10000 | 10000 |
| Observed class support | {'ER_negative': 34, 'ER_positive': 67} | {'ER_negative': 45, 'ER_positive': 42} |
| Platform | GPL570 | GPL570 |
| Measurement units/type | microarray_log2_normalized | microarray_log2_normalized |
| Deposited processing | GEO deposited log2 GC-RMA signals; no additional normalization | GEO deposited RMA Bioconductor default parameters and log2 transformation; no additional normalization |
| Independence declaration | unknown | unknown |

▶ Complete task checks and input declarations

## Source screen and frozen target family

| Quantity | Count |
|---|---|
| Source features screened | 10000 |
| Source BH q ≤ 0.05 | 1339 |
| Frozen selected genes | 20 |
| Testable in target | 20 |
| Matching + target BH support | 14 |
| Matching without sufficient support | 5 |
| Opposite estimates | 1 |
| Untestable | 0 |
| Direction denominator | 20 |

Contrast: {'ER_negative': 'Clinically ER-negative primary breast tumor', 'ER_positive': 'Clinically ER-positive primary breast tumor'}. Signed effect is positive-group mean minus negative-group mean in each cohort's deposited normalized log-intensity units. Absolute source/target scales are not assumed comparable.

Source BH correction covers the entire screened family; target BH correction covers every frozen hypothesis, including missing tests as p=1 for correction and displayed q=null. Matching without support is inconclusive; opposite signs alone do not establish effect reversal.

Frozen hypothesis SHA-256: `8f17b0e452532a9f0add71d4ce96986315d84a609c709d73c3e35890e04f446f`. Source-only selection settings: {'max_genes': 20, 'rank': 'descending absolute Hedges g, then feature ID', 'source_bh_q_max': 0.05}.

## Every selected association, including threshold-not-met and unavailable outcomes

| Rank | Feature ID | Source effect | Source p | Source q | Target effect | Target p | Target q | Direction | Evidence / reason |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 135112 | -1.5656843680992107 | 6.606687896479625e-11 | 6.606687896479625e-07 | -0.5915192438634893 | 0.0019810642785575154 | 0.005660183653021472 | True | supported_matching_direction |
| 2 | 7091 | -1.2123939866935913 | 1.643178310589727e-09 | 8.215891552948635e-06 | -0.3956866160746042 | 0.006020286098264572 | 0.010945974724117404 | True | supported_matching_direction |
| 3 | 29968 | -2.504760134847674 | 1.1705102832788857e-07 | 0.00015072860036690158 | -2.434493945692064 | 1.2000443168994843e-07 | 2.4000886337989688e-06 | True | supported_matching_direction |
| 4 | 56935 | -1.7320985613582085 | 3.538138890966689e-09 | 8.845347227416722e-06 | -0.7800428881476211 | 3.451915659361692e-07 | 3.0792434547195223e-06 | True | supported_matching_direction |
| 5 | 10477 | -1.7049495736690101 | 2.53020227901173e-09 | 8.434007596705767e-06 | -0.7609911575269841 | 1.820175430299317e-05 | 9.100877151496584e-05 | True | supported_matching_direction |
| 6 | 140885 | -1.2849330898485531 | 1.7148304296876163e-06 | 0.0004512711657072674 | -0.3294232904158756 | 0.007382750937810231 | 0.012304584896350385 | True | supported_matching_direction |
| 7 | 5336 | -1.4769432295640907 | 4.565032493197977e-07 | 0.00025361291628877647 | -0.6385266318285723 | 0.0022797182784417016 | 0.005699295696104254 | True | supported_matching_direction |
| 8 | 5621 | -1.408887546945122 | 6.3357396437549594e-09 | 1.267147928750992e-05 | -0.3967059930015857 | 0.009787535935749887 | 0.01398219419392841 | True | supported_matching_direction |
| 9 | 84441 | -1.53074354409131 | 2.3373098975134298e-07 | 0.0001797930690394946 | -0.4413349670650808 | 0.009494487650521029 | 0.01398219419392841 | True | supported_matching_direction |
| 10 | 55816 | -1.550158714732221 | 3.1467611808754485e-05 | 0.0024584071725589443 | -0.16886908899682318 | 0.3709647300401912 | 0.4121830333779902 | True | matching_direction_without_support |
| 11 | 5997 | -1.5419161381782267 | 3.4866735635892614e-07 | 0.0002198758900780919 | -0.32713889236983995 | 0.13419202270619823 | 0.17892269694159765 | True | matching_direction_without_support |
| 12 | 4067 | -1.618448105134327 | 1.5653518752565659e-07 | 0.00015276751397267892 | -0.47909278845079495 | 0.0037119186290388035 | 0.008248708064530673 | True | supported_matching_direction |
| 13 | 10402 | -1.4398429128854273 | 6.901011597370724e-06 | 0.0010101367783978338 | -0.2024709523206356 | 0.20751886965199348 | 0.2441398466494041 | True | matching_direction_without_support |
| 14 | 54947 | -1.105397366864354 | 3.4873315326805203e-06 | 0.0007265274026417751 | 0.25492376222222024 | 0.1434853893477173 | 0.1793567366846466 | False | opposite_direction |
| 15 | 157506 | -1.7566425173968394 | 1.3676557771532736e-05 | 0.0015196175301703 04 | -1.382056932206349 | 4.618865182079283e-07 | 3.0792434547195223e-06 | True | supported_matching_direction |
| 16 | 51191 | -1.5464366724416152 | 2.184653844489341e-07 | 0.0001797930690394946 | -0.1646639437730153 | 0.4985016159246367 | 0.5247385430785649 | True | matching_direction_without_support |
| 17 | 834 | -1.6125131128612828 | 7.282016006219228e-07 | 0.00029699094432303086 | -0.08977279632539492 | 0.6383149397296727 | 0.6383149397296727 | True | matching_direction_without_support |
| 18 | 29015 | -1.1797020130175602 | 4.6432533806874985e-06 | 0.0008291523894084818 | -0.5529787691698402 | 0.0012447791607018653 | 0.004149263869006218 | True | supported_matching_direction |
| 19 | 5918 | -2.613861278781826 | 1.5796668302762965e-06 | 0.00044031458499384636 | -1.067867543080954 | 0.005731681804330497 | 0.010945974724117404 | True | supported_matching_direction |
| 20 | 8424 | -1.9275273564784894 | 4.16426330018243e-05 | 0.0028522351371112536 | -1.6597479584365074 | 5.5159670511596764e-05 | 0.00022063868204638708 | True | supported_matching_direction |

None means unavailable, never an assumed zero. Exact source/target group sizes, Hedges' g, pointwise intervals and reasons remain in race/*.json and CSV. Pointwise intervals are not simultaneous or adjusted for source selection. No model transfer is established.

## Warnings and restrictions

- **PROCESSING_DIFFERENCE** · The cohorts declare different processing methods; shared measurement types do not guarantee directly comparable scales
- **TARGET_INDEPENDENCE_UNKNOWN** · Independent cohort acquisition is not documented for both cohorts
- **PATIENT_IDENTIFIERS_ABSENT** · Patient IDs are absent in one or both cohorts; cannot rule out overlapping subjects using identifiers alone
- **EXPLORATORY_ONLY** · Acquisition, patient independence and phenotype equivalence are declarations, not authenticated facts. No locked or clinical validation claim.

## Methods, provenance, exclusions and reproduction

See analysis_methods.json, compatibility_summary.json and warning_exclusion_records.json for full recorded context. Exclusions before uploaded files are unknown unless a verified preparation record is supplied. No imputation or silent exclusion is added by this export.

Read REPRODUCE.md and PRIVACY.md. Input file hashes identify original measurements; raw data must be acquired separately. CSV_TEXT_ENCODING.json records spreadsheet-safe text encoding without changing JSON identifiers. FILE_SHA256_MANIFEST.csv and reproducibility_manifest.json cover the published evidence.

## Consistency audit


```
{
  "status": "PASS",
  "checks": {
    "race": {
      "source_screened": 10000,
      "frozen": 20,
      "target_testable": 20,
      "supported": 14,
      "untestable": 0,
      "hypothesis_hash": "8f17b0e452532a9f0add71d4ce96986315d84a609c709d73c3e35890e04f446f"
    }
  },
  "note": "Independent count/formula/CSV/JSON/freeze audit; no model refit, threshold tuning or replacement of scientific estimates."
}
```


## Inspect the enclosed evidence

- CSV_TEXT_ENCODING.json
- FILE_SHA256_MANIFEST.csv
- PRIVACY.md
- PUBLIC_DATA_PREPARATION.md
- REPRODUCE.md
- analysis_methods.json
- compatibility_summary.json
- consistency_audit.json
- race/FILE_SHA256_MANIFEST.csv
- race/race_methods.json
- race/race_result.json
- race/replication_genes.csv
- race/selected_signature.json
- race/source_gene_screen.csv
- race_config.json
- reproducibility_manifest.json
- warning_exclusion_records.json